

Reading Reports

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Due Date	In-Text Citation (APA/MLA)	Key Term/Idea	Type of Source
9/26/25	AI diagnoses diseases with Less Data. (2019, April). Machine Design, 91(4), 18+. https://link.gale.com/apps/doc/A654037610/SUIC?u=atholtonhsmc&sid=bookmark-SUIC&xid=4f92c8b1	New algorithm used to identify disease	Article
9/26/25	AI Detects Early Signs of MCI and Dementia in Older Drivers. (2023, August). Healthy Years, 20(8), 2. https://link.gale.com/apps/doc/A757711877/SUIC?u=atholtonhsmc&sid=bookmark-SUIC&xid=2a5e11fa	Early detection of cognitive diseases with AI	Brief Article
10/10/25	Huhulea, E. et al. (2025). Artificial intelligence advancements in oncology: A review of current trends and future directions. Biomedicines, 13(4), 951. https://doi.org/10.3390/biomedicines13040951	Advancements in and challenges of using AI in Oncology	Scholarly Journal Article
10/10/25	Howard, F. M., et al. (2024). Generative adversarial networks accurately reconstruct pan-cancer histology from pathologic, genomic, and radiographic latent features. Science Advances, 10(27), Article eadq0856. https://doi.org/10.1126/sciadv.adq0856	How AI uses a tool to analyze data to predict cancer.	Scholarly Journal Article
10/24/25	Chua, I. S., Gaziel-Yablowitz, M., Korach, Z. T., Kehl, K. L., Levitan, N. A., Arriaga, Y. E., Jackson, G. P., Bates, D. W., & Hassett, M. (2021). Artificial intelligence in oncology: Path to	How AI is growing and being implemented in the field of Oncology	Scholarly Journal Article

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	implementation. Cancer Medicine, 10(12), 4138–4149. https://doi.org/10.1002/cam4.3935		
10/24/25	Hekler, A., Utikal, J. S., Enk, A. H., Hauschild, A., Weichenthal, M., Maron, R. C., Berking, C., Haferkamp, S., Klode, J., Schadendorf, D., Schilling, B., Holland-Letz, T., Izar, B., von Kalle, C., Fröhling, S., & Brinker, T. J. (2019). Superior skin cancer classification by the combination of human and artificial intelligence. European Journal of Cancer, 120, 114–121. https://doi.org/10.1016/j.ejca.2019.07.019	benefits of fusing human expertise and with artificial intelligence	Scholarly Journal Article.
11/7/25	Chen, C., Isa, N. A. M., & Liu, X. (2025). A review of convolutional neural network based methods for medical image classification. Computers in Biology and Medicine, 185. https://doi.org/10.1016/j.compbiomed.2024.109507	How Convolutional Neural Networks fare in image recognition.	Scholarly Journal Article.
11/7/25	Achir, A., Debbarh, I., Zoubir, N., Battas, I., Medromi, H., & Moutaouakkil, F. (2025). Advances in leukemia detection and classification: A systematic review of AI and image processing techniques. F1000Research, 13(1536). https://doi.org/10.12688/f1000research.159318.2	How AI is used in Leukemia detection	Scholarly Journal Article.
11/14/25	Bernstein, M. H., Atalay, M. K., Dibble, E. H., Maxwell, A. W. P., Karam, A. R., Agarwal, S., Ward, R. C., Healey, T. T., & Baird, G. L. (2023). Can incorrect artificial intelligence (AI) results impact radiologists,	How AI has the potential to negatively impact detection.	Scholarly Journal Article.

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	and if so, what can we do about it? A multi-reader pilot study of lung cancer detection with chest radiography. <i>European Radiology</i> , 33, 8263–8269. https://doi.org/10.1007/s00330-023-09747-1		
11/14/25	Bacha, A., & Shah, H. H. (2024). AI-Enhanced Liquid Biopsy: Advancements in Early Detection and Monitoring of Cancer through Blood-based Markers. <i>Global Journal of Universal Sciences</i> , 1(2), 68–86. https://doi.org/10.70445/gjus.1.2.2024.68-86	AI detection of cancer through liquid biopsy analysis	Scholarly Journal Article
11/21/25	Al-Rawi, N., Sultan, A., Rajai, B., Shuaeeb, H., Alnajjar, M., Alketbi, M., Mohammad, Y., Shetty, S. R., & Mashrah, M. A. (2022). The effectiveness of artificial intelligence in detection of oral cancer. <i>International Dental Journal</i> , 72(4), 436–447. https://doi.org/10.1016/j.identj.2022.03.001	Effectiveness of AI in detecting oral cancer	Scholarly Journal Article

AI Detects Early Signs of MCI and Dementia in Older Drivers. (2023, August). *Healthy Years*, 20(8), 2.

<https://link.gale.com/apps/doc/A757711877/SUIC?u=atholtonhsmc&sid=bookmark-SUIC&xid=2a5e11fa>

Summary: In this Research Article, Columbia University researchers discuss how an algorithm they developed predicts disease in older adults through their driving habits. Through the usage of a large driving study, the model that they were able to create achieved a 96% accuracy rate in predicting mild cognitive impairment and dementia, which outperformed other machine learning models. Two of the most influential factors in these diagnoses are the right to left turn ratio, along

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with the total incidents of hard breaking, which correlate with advanced age. Effectively predicting mild cognitive impairment and dementia helps to intervene and prevent any future issues, especially in the absence of therapeutics.

Application to Research:

This study demonstrates the ability of AI to detect and predict the onset of a mental disease, which can be hard to predict normally, just through the analysis of data, which allows it to correlate certain behaviors with mental decline. By showing this capability of AI,

AI Diagnoses Diseases with Less Data. (2019, April). Machine Design, 91(4),

<https://link.gale.com/apps/doc/A654037610/SUIC?u=atholtonhsmc&sid=bookmark-SUIC&xid=4f92c8b1>

Summary:

Despite the abilities of AI becoming better and better at recognition and classification of visual data, they are still unlikely to replace human doctors, since they still need data labelled by humans, and large amounts of it. To advance AI further, Carnegie Mellon Researchers developed a learning technique called MedAl(medical active learning) that requires less labelled images but still achieves similar results to conventional methods. The algorithm does this by labelling a set of images, plotting them, adding them to the data set, and making sure that the distribution is preserved. Tests showed that MedAL achieved an accuracy of 80% when detecting diabetic retinopathy while using 32% less labelled images. It was also showcased to work on other diseases like skin and breast cancer, giving an example of how it can be applied to a variety of problems that have data constraints.

Application to Research:

The research done by the Carnegie Mellon researchers shows how AI can be continuously improved to make it better by focusing the data itself. It addresses how AI has its limitations, but

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through efforts to make it better and more-cost efficient, it becomes a more viable option in aiding doctors to detect disease.

Huhulea, E. et al. (2025). Artificial intelligence advancements in oncology: A review of current trends and future directions. *Biomedicines*, 13(4), 951. <https://doi.org/10.3390/biomedicines13040951>

Summary

This journal article from NIH explains how Artificial Intelligence innovations and their potential in helping identify cancer, treat it, and manage it. Through surveying the many different ways that AI is being utilized, such as in nanomedicine and immunotherapy, and comparing them to traditional methods, it was clear that using and being assisted by the models could be much better than traditional ways of doing things. However, it is difficult to make AI methods more popular due to difficulties in things such as the data quality used to train the models, bias, and more.

Application to Research

This article gives a really thorough overview of Artificial Intelligences applications in medicine, particularly in the field of oncology. It provides important info about how exactly this is done, which is relevant to my topic of how artificial intelligence is used to diagnose and treat cancer. As this is a very long research article which contains many different pieces of background information, such as how these AI models can be flawed, it will be useful.

Howard, F. M., et al. (2024). Generative adversarial networks accurately reconstruct pan-cancer histology from pathologic, genomic, and radiographic latent features. *Science Advances*, 10(27), Article eadq0856. <https://doi.org/10.1126/sciadv.adq0856>

Summary

This journal article details a new way that AI can be used to accurately predict disease after being trained with non-conventional data sources, a generative adversarial network, which essentially allows it to see what is wrong at a cellular level without having to physically cut people open for

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samples to analyze. Through evaluating the new tool over many different categories of cancer, it demonstrated an ability to accurately discern information like tumor grade, its subtype, and its gene expression patterns, all valuable things when it comes to treating cancer. With the development of the Generative adversarial network, it isn't needed to invasively examine individuals for cancer, and improve the accuracy of cancer diagnoses and treatments.

Application to Research

This article effectively showcases an innovation in the field of Oncology using generative AI to create images of a person's tissue from scans, which prevents from having to actually having to cut out a piece of someone to figure out what's wrong. It shows the wide application of AI, not only as a tool for analyzing and doing everything on its own, but also working in conjunction with healthcare professionals to do things more accurately and efficiently, in this case, with generative artificial intelligence. This will be useful for discussing how innovations play a part in my topic.

Chua, I. S., Gaziel-Yablowitz, M., Korach, Z. T., Kehl, K. L., Levitan, N. A., Arriaga, Y. E., Jackson, G. P., Bates, D. W., & Hassett, M. (2021). Artificial intelligence in oncology: Path to implementation. *Cancer Medicine*, 10(12), 4138–4149. <https://doi.org/10.1002/cam4.3935>

Summary

In this article, the ways in which AI is able to change cancer care and be implemented is explored. AI development has seen a huge growth, especially in their usage in the field of Oncology. Specifically, deep learning and machine learning models have shown promise in cancer imaging, where it can analyse tissue, CT scans, and MRIs to detect tumors, along with predicting patient outcomes, such as recurrence and mortality. In many different stages of cancer treatment, improvements have been seen through the usage of AI, such as it being more accurate in detection, being able to “notice” things that a human might not notice, better tailor treatments to individuals patients, optimize response, and helping to reduce cost and expand access to patient care, since it

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has the ability to provide guidance on the level of experts, and optimize resource allocation.

However, there are many barriers to implementing Artificial Intelligence in the field of Oncology and enjoying the benefits that it provides. One of its main issues is that the data used to train artificial intelligence can be fragmented, unstandardized, and biased, making it difficult to create models, and if the bias is not taken care of, it can be reflected in the model, leading to inaccurate predictions/diagnoses. This is a big problem, as if you fully supplement for a human, someone could suffer direct consequences as a result of either remaining undiagnosed, receiving the wrong type of treatment, etc. Furthermore, people may be very unfamiliar with using artificial intelligence, making it difficult to implement in actual settings. In order to address these issues, there are several strategies that could be used for further implementation: educating/training healthcare professionals about AI, rigorously testing algorithms and models, and developing better frameworks for implementation.

Application to Research:

This article will be very good for my research because it addresses what AI is capable of, while also discussing what limits and barriers it currently has. It provides a highly detailed description of how AI is being improved in the field of oncology, such as with detection through image analysis, and creating treatments, but by also detailing what issues it faces in implementation, it will help me explain that AI can be effective, but only so much in the real world.

Hekler, A., Utikal, J. S., Enk, A. H., Hauschild, A., Weichenthal, M., Maron, R. C., Berking, C.,

Haferkamp, S., Klode, J., Schadendorf, D., Schilling, B., Holland-Letz, T., Izar, B., von Kalle, C.,

Fröhling, S., & Brinker, T. J. (2019). Superior skin cancer classification by the combination of human and artificial intelligence. *European Journal of Cancer*, 120, 114–121.

<https://doi.org/10.1016/j.ejca.2019.07.019>

Summary:

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This research article discusses how combining the expertise of human professionals with artificial intelligence algorithms can improve the accuracy of skin cancer classification. It details how in a study, 112 dermatologists with a convolutional neural network analyzed over 11000 images of various skin lesions, consisting of things like melanoma, basal cell carcinoma, and benign tumor growths. A gradient boosting model, in which multiple decision trees are trained consecutively, each fixing the previous's mistakes, was used to merge the results of the human system and the results of the artificial intelligence after analyzing the images of skin lesions. A different type of AI was required to merge the results since the AI was more objective than the experts. The combination of results, taking in the doctors' confidence levels and the neural network's output probability, led to an accuracy of 82.95%, which was higher than the neural network, and significantly higher than the dermatologists. It was also more sensitive in identifying malignant and benign tumors. The relative success of the combination demonstrates that collaboration between humans and AI produces far better results than both alone, which is especially true when detecting skin cancer early. This could be applied to other fields beyond dermatology, but it requires further testing before wide-spread use is adopted.

Application to Research:

This article is reliable and can be applied to my research because it effectively highlights a more effective way of using AI in the diagnosis of cancer, more specifically skin cancer. It details a study that shows through combining the confidence values of humans and AI, better accuracy when diagnosing and distinguishing between various cancers can be achieved. Even though it only focuses on a specific type of cancer, this application has the possibility of being applied further beyond the scope of just dermatologic cancers, showing that AI can be effective, and even more effective through a slightly different usage.

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Chen, C., Isa, N. A. M., & Liu, X. (2025). A review of convolutional neural network based methods for medical image classification. *Computers in Biology and Medicine*, 185.

<https://doi.org/10.1016/j.compbiomed.2024.109507>

Summary: This scholarly article analyzes 149 research papers on how convolutional neural networks are used to classify medical images such as X-rays, CT scans, and MRIs. The authors describe how CNNs(convolutional neural networks) automatically extract image features, improving disease detection accuracy and efficiency compared to older methods that relied on manually chosen image characteristics. The paper reviews CNN techniques such as data preprocessing, transfer learning, architecture design, and explainability, through the compilation of 38 public medical datasets. While CNNs have greatly advanced AI-assisted diagnosis across diseases like cancer, heart disease, and brain disorders, clinical implementation is still limited by small or imbalanced datasets, poor model implementation, and lack of transparency on its uses and benefits.

Application to Research:

This article will allow me to explain a specific way that AI is able to effectively improve cancer diagnosis and treatment, through convolutional neural networks. This type of AI is used in many things, such as in CT scans, MRIs, X-rays, and tissue slides, and identifies tumors through image recognition. While it also outlines a clear way in which image recognition AI through convolutional neural networks can be hugely beneficial to the field of oncology, it also discusses major drawbacks that have an effect on how AI can be implemented.

Achir, A., Debbarh, I., Zoubir, N., Battas, I., Medromi, H., & Moutaouakkil, F. (2025). Advances in leukemia detection and classification: A systematic review of AI and image processing techniques. *F1000Research*, 13(1536). <https://doi.org/10.12688/f1000research.159318.2>

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Summary: Leukemia is a diverse group of cancers characterized by the abnormal proliferation of white blood cells, which impairs normal immune functions and blood circulation. The challenge that it poses is that it can be very complex, having many risk factors and variable outcomes. In the study, 2500 articles were reviewed in order to review how AI is being used in acute lymphoblastic leukemia, acute myeloid leukemia, chronic lymphocytic leukemia, and chronic myeloid leukemia. As a result, they were able to conclude the key risk factors for leukemia, such as chromosomal abnormalities, being exposed to toxins, and one's socioeconomic status. Artificial Intelligence models, especially convolutional neural networks, were able to accurately and reliably diagnose Leukemia based on all of the factors listed, also in conjunction with the other data such as images of blood smears and slides of bone marrow. AI offers a chance to drastically improve leukemia treatment, through early detection, more precise classification, and treatment planning. However, in order for these benefits to be seen, data has to be standardized, and access needs to be widespread.

Application to Research:

This source is important because it focuses on a single type of cancer, Leukemia, which will allow me to better introduce a more specific example in my argument of how effective AI is. It also introduces another way in which AI aids in detection, through analysis of risk factors, which will allow it to better make a decision. Like others, it also lists drawbacks and limitations of using AI, including inconsistent data and lack of access.

Bernstein, M. H., Atalay, M. K., Dibble, E. H., Maxwell, A. W. P., Karam, A. R., Agarwal, S., Ward, R. C., Healey, T. T., & Baird, G. L. (2023). Can incorrect artificial intelligence (AI) results impact radiologists, and if so, what can we do about it? A multi-reader pilot study of lung cancer detection with chest radiography. *European Radiology*, 33, 8263–8269.
<https://doi.org/10.1007/s00330-023-09747-1>

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Summary:

In this article, the effect of AI feedback and decision-making on Radiologist diagnostics in the interpretation of chest x-rays is examined. It discusses a study, where radiologists reviewed many x-rays across multiple different conditions, where usage of AI feedback varied from no AI to full on detection. They found that when the AI provided incorrect info, the error rate in identifying cancer significantly increased when compared to the condition where no AI was used. To mitigate this, researchers found that strategies like visual outlines of AI suggestions or deleting fully AI results from records, helped reduce the error rate.

Application to Research:

This article is important to my research because unlike many other articles, it specifically discusses a scenario where AI has been used, and how it impacts lung cancer diagnosis. While AI can support detection, it can harm the accuracy of actual humans when depending on it to help in identifying tumors. With these points, I will be better able to assess AI's general effectiveness in detection of cancer.

Bacha, A., & Shah, H. H. (2024). AI-Enhanced Liquid Biopsy: Advancements in Early Detection and Monitoring of Cancer through Blood-based Markers. *Global Journal of Universal Sciences*, 1(2), 68–86. <https://doi.org/10.70445/gjus.1.2.2024.68-86>

Summary:

This article is about how AI is being integrated into a non-invasive blood test, known as liquid biopsy, to better improve cancer diagnoses and treatment. Through the application of machine learning algorithms, liquid biopsies are better able to find biomarkers of cancer, enabling cancer to potentially be detected early. Usage of machine learning has allowed doctors to give more quality treatment of cancer and more personalized treatment plans. It also improves accuracy, identifying markers that might be missed otherwise. However, the issues of data quality used for training AI,

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the model's accuracy, and integration clinical settings need to be solved in order for AI-enhanced liquid biopsy to transform cancer care.

Application to Research:

This article provides detailed information on another way in which AI can help in detecting cancer, through enhancement of liquid biopsies. This will better allow me to create an argument where I include not just image recognition and data analysis, but active integration into a tool that doctors normally use to screen for cancer.

Al-Rawi, N., Sultan, A., Rajai, B., Shuaeeb, H., Alnajjar, M., Alketbi, M., Mohammad, Y., Shetty, S. R., & Mashrah, M. A. (2022). The effectiveness of artificial intelligence in detection of oral cancer. *International Dental Journal*, 72(4), 436–447. <https://doi.org/10.1016/j.identj.2022.03.001>

Summary:

Oral cancer is the 11th most reported cancer worldwide, and despite major improvements in treatment, its mortality rate is still high, leading to consideration of an increase in the use of AI to help diagnose it early. Diagnosing it early allows for significant improvements in the rates of morbidity and mortality. A total of 17 studies were conducted, with 6 using deep learning, and the remaining 11 using supervised machine learning. For deep learning, which used models such as Convolutional neural networks, they performed relatively the same as the supervised machine learning models. Deep learning was more accurate, but both hold promise for the detection of Oral cancer.

Application of Research:

The article discusses AI usage in a specific type of cancer, oral cancer, and it compares two different examples of Artificial Intelligence, and their usage in the early detection of oral cancer. By providing direct comparison of deep learning and supervised machine learning, I will be able to better provide details on AI itself, and not just its role and effectiveness.

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